



Do monitoring or advisory lead independent directors with financial expertise matter to managerial opportunism?

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ABSTRACT

Motivated by the increasing interest and the monitoring and advisory roles that a lead independent director may play to decrease managerial opportunism; we examine the relationship between monitoring or advisory lead independent directors and managerial opportunism proxied by earnings management. Using a panel data of US firms from 2001 to 2020, we find that monitoring lead independent directors rather than advisory lead independent directors is negative and significantly associated with discretionary accruals. In an additional analysis, we find that advisory rather than monitoring lead independent directors improve earnings persistence and future cash flows, suggesting that both monitoring and advisory lead independent directors play complementary roles when it comes to corporate performance/earnings quality. We also find that, regardless of the role they play within boardrooms, monitoring and advisory lead independent directors with financial expertise have a decreasing effect on discretionary accruals. These findings provide new insights and improve our understanding of how monitoring and advisory lead independent directors contribute to corporate performance in complementary ways.

1. Introduction

With the ongoing discussions on board leadership structure and the pressure from corporate watchdogs and legislators to separate the Chief Executive Officer (CEO) and the board chair positions, regulators are facing pressure to implement a lead independent director policy. The appointment of a lead independent director extends beyond addressing the concerns associated with the combined role of the CEO and the board chair (CEO duality). For instance, while the United States (US) regulators in an attempt to resolve the concerns raised by corporate watchdogs and legislators, moved to implement a lead independent director policy to avoid the separation of the CEO and the board chair positions (Dalton & Dalton, 2005; Krause et al., 2017), regulators in the European countries including France, Germany, Italy and the United Kingdom (UK), where CEO duality is rare, have also implemented similar policy (Davies & Hopt, 2013; Allianz Global Investors, 2022). Nevertheless, these regulators have not made it clear whether lead independent directors should play one of the two main documented board

of directors' roles – monitoring or advisory (Adams & Ferreira, 2007; Faleye et al., 2013; Hsu & Hu, 2016)¹ – or whether it is sufficient for these directors to merely add to the board leadership structure.

According to Krause et al. (2017), a lead independent director is appointed to counterbalance the CEO's power. Indeed, academics and regulators contend that a lead independent director representation on the board (i) should be considered as a credible alternative when CEO duality is visible, (ii) provides a sounding board for the chair and serves as a go-between the remaining directors and shareholders, (iii) attends to the concerns and builds consensus when there is a dispute between the CEO and the board chair, and (iv) ensures investors' concerns that have not been addressed by the board are resolved (Owusu et al., 2023). Consequently, a considerable number of studies have documented the positive impact of a lead independent director on multiple corporate outcomes including corporate governance quality, firm performance, managerial risk-taking, and cost of debt (Krause et al., 2017; Lamoreaux et al., 2019; Rajkovic, 2020; Owusu et al., 2023). More pertinent to our study, though, is that Bryan and Mason (2023) and Hsu et al. (2024)

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¹ It is important to highlight that those directors assigned to perform monitoring roles serve on audit, governance, nomination, and compensation committees, while those who serve on strategy, investment, and finance committees perform advisory roles (Faleye et al., 2011, 2013; Kim et al., 2014).

show that lead independent directors' representation on the board has the propensity to reduce managerial opportunism associated with earnings management, and this is especially so when they serve on the audit committee, become older, serve longer on the board and when they have a higher educational background (Ph.D.). Their evidence, however, is in sharp contrast to Larcker et al. (2007) who find no association between a lead independent director representation on the board and earnings management.

Nevertheless, existing regulators and academic researchers precisely focused on the mere addition of a lead independent director to the board leadership structure without considering the channels through which a lead independent director may contribute to the board leadership effectiveness. For instance, the mere appointment of a lead independent director appears to be a symbolic management tactic to satisfy regulators with less effectual (Shi & Connelly, 2018). However, a lead independent director policy has been influenced by the concerns raised by corporate watchdogs and legislators without being informed by reliable evidence associated with their potential impact on board leadership effectiveness (Dalton & Dalton, 2005). In terms of the board leadership roles, it is not empirically clear in existing literature whether the monitoring role or advisory role is most suitable for lead independent directors. So far, no study has gone beyond the mere participation of lead independent directors on corporate boards and examined how these directors' monitoring or advisory roles affect earnings management. Our study fills this gap in the existing literature by examining the differential effect of monitoring or advisory lead independent directors on earnings management.

Additionally, prior research (e.g., Bryan & Mason, 2023; Hsu et al., 2024) aimed at exploring the underlying mechanisms through which the mere participation of lead independent directors on the board affects earnings management has focused on their experience and expertise such as audit committee membership, age, tenure and educational background with insufficient attention to their financial expertise. Nevertheless, to use their capacity as better monitors or advisors to mitigate the financial reporting decisions that violate the underlying accounting principles, lead independent directors may need to have a better understanding of the financial reporting process, and therefore their financial knowledge is crucial. Our study fills this gap in the existing literature by examining the role of financial expertise in the relationship between monitoring or advisory lead independent directors and earnings management.

There are several reasons that a lead independent director monitoring role rather than an advisory role can enhance corporate governance and constrain earnings management. First, in common with the compromise board leadership structure theory which assumes that the appointment of a lead independent director allows managerial autonomy with the required monitoring (Krause et al., 2017), we argue that the monitoring rather than the advisory role of a lead independent director is likely to exert influences on managers to adopt financial reporting decisions in favour of less earnings management. Second, unlike the lead independent directors in an advisory role, the membership in monitoring committees (e.g., audit committee) allows lead independent directors in a monitoring role to have direct interaction with the external audit firm and challenge a powerful CEO and managers to address accounting issues raised by the committees and the external auditors. Therefore, one can argue that, because of lead independent directors' unique responsibilities and power, their monitoring rather than advisory role is more likely to play a crucial role when it comes to the quality of financial reporting. Third, prior literature documents that the opportunistic behaviour of managers can be controlled through monitoring (see Wild, 1996; Dechow et al., 1996), so a lead independent director in a monitoring role than an advisory role can improve the credibility, integrity, and reliability of financial information. We, thus, expect a lead independent director in a monitoring, rather than an advisory role, to be associated with a decrease in earnings management.

To test our prediction, we use discretionary accruals as a measure of

managerial opportunism. Our empirical analysis employs a panel dataset consisting of 6,664 US firm-year observations between 2001 and 2020. We find strong evidence to support our prediction that a lead independent director in a monitoring role, rather than in an advisory role, is negatively and significantly associated with discretionary accruals. Our findings continue to hold after using a number of robustness tests including alternative measures of discretionary accruals, monitoring and advisory lead independent directors, and several techniques to control for endogeneity.

Next, we examine whether a lead independent director's monitoring and advisory roles may complement each other when it comes to corporate performance. While Hsu and Hu (2016) demonstrate that independent directors' advisory role rather than their monitoring role is more effective in increasing earnings persistence, our findings suggest that a lead independent director monitoring role rather than an advisory role reduces abnormal accruals. These contrasting findings may be due to the difference in earnings quality measures used and calls for further tests to explore whether lead independent directors' monitoring and advisory roles contribute to corporate performance in complementary ways. We show that firms with a lead independent director monitoring role are less likely to benefit from persistent earnings quality and higher future cash flow than those firms with a lead independent director in an advisory role. These results suggest that lead independent directors' monitoring and advisory roles contribute to corporate performance in complementary ways.

In addition, we assess the underlying mechanism through which a lead independent director may use to reduce discretionary accruals in their monitoring or advisory roles. Specifically, we explore how financial expertise moderates the relationship between a lead independent director monitoring or advisory roles and earnings management. We show significant results that lead independent directors in monitoring or advisory roles with financial expertise can restrain managerial opportunism toward discretionary accruals. Our findings suggest that lead independent directors with financial expertise have the propensity to decrease earnings management regardless of the role that they play within the boardroom.

Our study advances the literature in two important aspects. First, we expand the literature on the relationship between lead independent directors and earnings management (Larcker et al., 2007; Bryan & Mason, 2023; Hsu et al., 2024) by analysing whether the monitoring or advisory lead independent director plays any role in the relationship. In doing so, we place each lead independent director in a specific role and treat the monitoring role than advisory role as the key determinant of earnings quality. Our novel approach draws on the two documented monitoring and advisory board roles (Adams & Ferreira, 2007; Faleye et al., 2011). These differentiated board roles form the basis for our discussion and interpretation of the results, enabling us to advance the line of research examining the novel contribution of the mere presence of a lead independent director on the board to a lead independent director role-earnings management relationship.

Second, we advance the literature on the relationship between the board of directors and earnings persistence (Hsu & Hu, 2016) by showing how monitoring and advisory lead independent directors play complementary roles. Unlike Hsu and Hu (2016) who find that advisory than monitoring independent directors increase earnings persistence, we reveal that while monitoring lead independent director role reduces abnormal accruals, their advisory role increases earnings persistence. These findings have positive implications for firms. To ensure strong oversight, accountability, and compliance, especially when it comes to management discipline around financial reporting quality, it is important for firms to appoint a lead independent director to a monitoring role, while in the areas of strategic insight and persistent earnings, an advisory lead independent director is suitable. We conclude that firms can benefit from earnings quality (earnings persistence) when they appoint lead independent directors and assign them to the monitoring role (advisory role).

The rest of the paper is organised as follows. The next section discusses the related literature and hypothesis development. [Section 3](#) presents data and explains how earnings management is constructed. [Section 4](#) provides the empirical analysis. [Section 5](#) concludes the paper.

2. Related literature and hypotheses

2.1. Theoretical framework

A number of theories, including agency, stewardship or classical organisation, and compromise board leadership structure theories, can broadly be used to predict the association between lead independent director roles and earnings management. For example, agency theory has long been used to study board leadership structure (e.g., [Berle & Means, 1932](#); [Jensen & Meckling, 1976](#)). This theory postulates that the inherent conflict of interests between managers and outside shareholders has long been established (e.g., [Berle & Means, 1932](#)) and, as a result, a board leadership structure with a combined CEO and the board chair positions has the propensity to empower one individual to take an entrenched position and increase agency costs ([Fama & Jensen, 1983](#)). Therefore, role separation between the CEO and the board chair positions arguably allows for checks and balances with the possibility of constraining managerial opportunism. [Fama \(1980\)](#) and [Jensen and Meckling \(1976\)](#), however, argue that because managers are anxious about their reputation and employment risk, they become risk-averse, and thus independent monitoring alongside incentives is needed to encourage managerial risk-taking to influence their underlying financial performance ([Baysinger & Hoskisson, 1990](#); [Jensen & Murphy, 1990](#)). These arguments lead to the possibility that the monitoring and advisory roles of lead independent directors may be needed to increase managerial risk-taking to boost the underlying financial performance but managers' tendencies toward risk-taking are likely to encourage them to engage in opportunistic incentives driving behaviour such as accruals earnings management.

Stewardship or classical organisation theory, however, argues that a board leadership structure with the combined CEO and board chair positions brings about the unity of command and clear lines of responsibility, thereby leading to quick decision-making ([Fayol, 1949](#); [Pfeffer, 1981](#); [Dalton et al., 1998](#)). The combined CEO and board chair positions arguably align managers' incentives with the objectives of the firm, and thus managers are less motivated to engage in economic private interests. [Donaldson and Davis \(1991; 1994\)](#) argue that these managers possess firm-specific knowledge and expertise necessary to implement suitable accounting policies that are likely to mitigate accruals earnings management and encourage them to provide accurate financial information for investors. These arguments suggest that a board leadership structure with a combined CEO and board chair positions is less likely to engage in managerial opportunism and, as a result, the monitoring and advisory roles of lead independent directors are less likely to make a difference.

In an attempt to reconcile the contrasting standpoints of agency theory and stewardship or classical organisation theory, [Krause et al. \(2017\)](#) developed the compromise board leadership structure theory, which believes that board leadership structure in the presence of a lead independent director provides a compromise solution to managerial autonomy and the necessary monitoring. They argue that the appointment of a lead independent director provides a balanced power, especially when the CEO and the board chair positions are occupied by one individual thereby constraining managerial opportunism. On the other hand, if there is a role separation between the CEO and the board chair positions but the CEO power is not sufficiently tempered by an independent board chair then the presence of a lead independent director is more likely to mitigate the opportunistic incentives driving behaviour. These arguments suggest that the monitoring and advisory roles of lead independent directors may be effective in mitigating managerial opportunism when there is a combined CEO and board chair position or

role separation between the two, thereby constraining accruals earnings management. In summary, existing board leadership structure theory provides competing predictions on how the monitoring and advisory roles of lead independent directors should impact managerial opportunism.

2.2. Empirical research

Several studies have attempted to analyse the relationship between a lead independent director's representation on the board and different corporate outcomes, but the results are less clear. [Chen and Ma \(2017\)](#) find that CEO dismissal due to poor firm stock performance is sensitive to the presence of a lead independent director. [Krause et al. \(2017\)](#) find that under the right conditions, the presence of a lead independent director improves firm performance, especially through a balanced power that is reflected in the board leadership structure. [Lamoreaux et al. \(2019\)](#) find that the appointment of a lead independent director is not window dressing because their presence on the board improves firm value and corporate governance quality. [Rajkovic \(2020\)](#) finds that a lead independent director representation on the board is more sensitive to investment efficiency and future firm performance. [Owusu et al. \(2023\)](#) report that the presence of a lead independent director decreases managerial risk-taking and the corresponding cost of debt. These findings are consistent with lead independent directors providing a compromise solution to managerial autonomy and the necessary monitoring to improve corporate outcomes. However, [Shi and Connelly \(2018\)](#) argued and provided evidence to show that the appointment of a lead independent director is a symbolic management tactic to satisfy regulators. They find that firms listed on the New York Stock Exchange (NYSE) compared to those listed on the National Association of Securities Dealers Automated Quotations (NASDAQ) appoint lead independent directors to deceive transient institutional investors than dedicated institutional investors because their appointment to meet the NYSE listing requirement relative to the organic adoption under the NASDAQ is less effectual.

While the foregoing empirical research is useful in terms of broadly highlighting the value of a lead independent director representation on the board, several important questions remain unresolved. First, existing research has mainly analysed the mere representation of lead independent directors on corporate boards and disregarded the differential effect of the two most important roles (i.e., monitoring and advisory) that each director on the board should perform. This distinction is extremely important because extant research (e.g., [Jensen, 1993](#); [Adams & Ferreira, 2007](#); [Faleye et al., 2011, 2013](#); [Hsu & Hu, 2016](#)) suggests that corporate boards perform two main roles – monitoring and advisory – and that independent directors tend to be more involved in one instead of the two main roles due to the limited time available to them. Second, whereas in the early 2000 s regulators in the US introduced the appointment of a lead independent director to avoid role separation between the CEO and the board chair (e.g., [Dalton & Dalton, 2005](#); [Krause et al., 2017](#)), it is uncertain whether these directors are appointed to perform monitoring or advisory roles that can be potentially more beneficial to shareholders. Alternatively, firms might be motivated to appoint these directors to perform non-strategic roles that are detrimental to shareholders to satisfy regulatory requirements.

Third, given that a lead independent director provides a compromise solution to ensure a balanced power is reflected in the board leadership structure ([Krause et al. 2017](#)); it is surprising that their impact on managerial opportunism measured in terms of earnings management has received limited attention. An exception to this is a recent study by [Bryan and Mason \(2023\)](#) and [Hsu et al. \(2024\)](#) who report that firms with a lead independent director representation on the board engage in less earnings management, but their evidence is in sharp contrast with an earlier work by [Larcker et al. \(2007\)](#) that report no association between a lead independent director and earnings management. While [Larcker et al. \(2007\)](#), [Bryan and Mason \(2023\)](#) and [Hsu et al. \(2024\)](#)

studies provide useful insights into the value of a lead independent director in constraining managerial opportunism, they simply focused on the mere participation of lead independent directors and disregarded the differential effect of these directors' main roles that they should play. Therefore, we seek to contribute to the existing literature by analysing the differential effect of lead independent directors' monitoring and advisory roles on managerial opportunism.

2.3. Hypotheses

The theoretical discussion above provides competing predictions on how a lead independent director's monitoring and advisory roles impact managerial opportunism and suggests the need to analyse the differential effect of lead independent directors' roles on accruals earnings management. The preceding empirical research broadly suggests that the mere participation of a lead independent director is related to a number of corporate outcomes. In appointing lead independent directors, corporate boards are expected to assign them to one of the two main board roles – monitoring and advisory. While independent directors who serve on audit, governance, nomination, and compensation committees are considered as undertaking monitoring roles, those who serve on strategy, investment, and finance committees engage in advisory roles (Faleye et al., 2011, 2013; Kim et al., 2014). Considering that the monitoring committees are dominated by independent directors who spend more time undertaking their monitoring responsibilities than the executive-dominated advisory committees (Heidrick & Struggles, 2007; Adams, 2009; Hsu & Hu, 2016), one can argue that lead independent directors in monitoring rather than advisory roles have the propensity to constrain managerial opportunism. This expectation is consistent with the compromise board leadership structure theory argument that the presence of a lead independent director provides a compromise solution to managerial autonomy and the necessary monitoring (Krause et al., 2017).

In addition, as the membership in monitoring committees allows lead independent directors in a monitoring role to interact with the external audit firm and the CEO when it comes to the financial reporting decisions compared with lead independent directors in an advisory role, we argue that this direct interaction between them is more likely to address financial reporting issues, suggestions, and concerns raised by the committees and the external auditors. An interesting parallel argument may be made here that their unique position in the board leadership structure suggests that lead independent directors when serving on monitoring committees are likely to strengthen the committees' power, improve the oversight of the financial reporting process, and effectively and efficiently mediate financial reporting issues between managers and external auditors. If firms with lead independent directors engage in less accruals earnings management as reported by Bryan and Mason (2023) and Hsu et al. (2024), then we would expect lead independent directors in monitoring rather than advisory roles to constrain managerial opportunism measured in terms of accruals earnings management. This leads us to our first hypothesis in an alternative form as follows:

H1: Accruals earnings management is lower for firms with a lead independent director in a monitoring role rather than in an advisory role.

Next, we attempt to examine whether our predicted relation between monitoring and advisory lead independent directors and accruals earnings management is sensitive to their specialist skills and professional expertise, and in our particular case, their financial expertise. This is important because extant studies (i.e., Xie et al., 2003; Abbott et al., 2004; Bedard & Johnstone, 2004; Dhaliwal et al., 2010; Badolato et al., 2014) suggest that independent directors with a financial background have the propensity to constrain earnings management. However, these studies have not always analysed how lead independent directors distinctive from regular independent directors with a financial background are related to earnings management, particularly in the case of

monitoring and advisory lead independent directors, which suggests a very important limitation. This is particularly so in the context of the argument that directors can be efficient monitors of managers if they have monitoring incentives and understand the ramifications of financial reporting decisions (Beekes et al., 2004).

Although Bryan and Mason (2023) and Hsu et al. (2024) find that the presence of lead independent directors conditioned by their experience and expertise (i.e., audit committee membership, age, tenure and higher educational background) is strongly associated with lower accruals earnings management than when not conditioning, they have not paid sufficient attention to how lead independent directors in the monitoring and advisory roles with financial expertise that is distinct from regular independent directors affects earnings management. We argue that the monitoring rather than advisory lead independent directors' ability to constrain earnings management largely depends on their monitoring incentives and a better understanding of the financial reporting process and the underlying accounting principles. Unlike regular independent directors who have less power and authority to challenge the CEO and managers, lead independent directors with additional leadership responsibilities among independent directors are better placed to have regular interaction with the officers and present any financial reporting issues raised by regular independent directors and auditors to be corrected, especially when they have financial expertise. Therefore, the financial expertise of lead independent directors is likely to strengthen their monitoring abilities to improve the quality of financial reporting.

As discussed, and hypothesise above, we would expect accruals earnings management to be lower for firms with a lead independent director in a monitoring role rather than in an advisory role. If monitoring than advisory lead independent director decreases accruals earnings management, then we would expect this result to be more prominent in firms that appoint monitoring rather than advisory lead independent directors with financial expertise. This leads us to our second hypothesis in an alternative form as follows:

H2: The negative relation between monitoring than advisory lead independent directors and accruals earnings management is stronger for firms that appoint lead independent directors with financial expertise.

3. Sample, data and variable definitions

3.1. Sample and data

We construct our sample by first collecting monitoring and advisory data from the BoardEx database, where the monitoring and advisory committees that the lead independent directors serve on, and other board-related variables are made available. Considering that the lead independent director's role became prominent in the early 2000 s, we begin constructing our sample from 2001. We also collect financial data required to calculate our dependent and control variables from the Thomson Reuters Worldscope database. We then merge the two distinct datasets and undertake the following procedures. First, we delete firm-year observations with missing data to calculate our independent, dependent and control variables. Second, we delete financial sector firms from our sample due to their distinct financial reporting and regulatory environment. Third, we delete industries with less than 20 industry-year observations in order to have sufficient data to calculate our dependent variable (i.e., earnings management). Finally, we winzorise all continuous variables at the 1st and 99th percentiles to reduce the potential outlier concerns. We obtain a total number of 6,664 firm-year observations over the period 2001–2020 after the preceding procedures to undertake our empirical analysis.

3.2. Proxy for earnings management

Following Dechow et al. (1995) and a more recent study by Sánchez-Ballesta and Yagüe (2022), we use the modified Jones model to capture

discretionary accruals (DA). Specifically, we use the following expectation model annually for each two-digit SIC industry-year with at least 20 industry-year observations:

$$\frac{TA_{it}}{A_{it-1}} = \alpha_0 + \delta_1 \left(\frac{1}{A_{it-1}} \right) + \delta_2 \left(\frac{\Delta Sales_{it} - \Delta Rec_{it}}{A_{it-1}} \right) + \delta_3 \left(\frac{PPE_{it}}{A_{it-1}} \right) + \varepsilon_{it} \quad (1)$$

$$DA_{it} = \frac{TA_{it}}{A_{it-1}} - NDA_{it}, \quad (2)$$

where in equation (1) TA_{it} captures total accruals, which is measured as the difference between net income before extraordinary items and operating cash flow. $\Delta Sales_{it}$ is a change in sales; ΔRec_{it} is a change in accounts receivable; PPE_{it} is the gross property, plant and equipment; A_{it-1} is lagged total assets. We estimate equation (1) cross-sectionally for each two-digit SIC industry-year with at least 20 observations and then use the residual to estimate the normal accruals for our sample firms. In equation (2), the DA_{it} (discretionary accruals) is the difference between total accruals and non-discretionary accruals (NDA_{it}).

Considering that a positive DA indicates income-increasing discretionary accruals and a negative DA indicates income-decreasing discretionary accruals, we follow Klein (2002) and more recent studies by Sánchez-Ballesta and Yagüe (2022), Owusu et al. (2022) and Hsu et al. (2024) and use the absolute value of DA to capture accruals manipulations in both directions. The higher the value of DA indicates greater the accruals manipulations and vice versa.

3.3. Proxy for monitoring and advisory lead independent directors

In identifying monitoring and advisory lead independent directors, we first follow previous literature (e.g., Bryan & Mason, 2023; Hsu et al., 2024; Owusu et al., 2025) and categorise each director as a lead independent director if the role title in BoardEx is lead director, lead independent director, lead outside director, presiding director, facilitating director, lead trustee, lead independent trustee or presiding trustee. We then measure monitoring and advisory lead independent directors following Faleye et al. (2011). Specifically, we consider an independent director to be in a monitoring role if he or she serves on not less than two of the four main monitoring committees such as audit, governance, nomination and compensation. We then extend this definition to lead independent monitoring directors ($LMONT$), which captures the percentage of these directors to the number of monitoring directors. As in Faleye et al. (2011), we capture advisory directors as those who do not serve on any of the monitoring committees. In light of this, we measure lead independent advisory directors ($LADV$) as a percentage of these

directors to the number of advisory directors that serve on committees other than audit, governance, nomination and compensation.

3.4. The measure of lead independent director with financial expertise

We capture lead independent directors with financial expertise ($LEADFE$) by using the BoardEx North America board and director committees file to identify lead independent directors with financial expertise. $LEADFE$ is coded 1 if a lead independent director committee role name has “Financial Expert”, and 0 otherwise.

4. Empirical analysis

4.1. Descriptive statistics

Table 1 summarises the descriptive statistics of the main variables employed in this study. While the mean percentage of monitoring lead independent directors ($LMONT$) is around 9% with a maximum value of 33, the advisory lead independent directors represent the average value of around 4 with the highest value of 20. The mean discretionary accruals (DA) in our sample is around 0.092. Table 2 also presents the Pearson correlation between the dependent and independent variables used in the main analysis. The results show that the correlation between DA and $LMONT$ is negative and a positive association between DA and $LADV$, indicating that monitoring lead independent directors have the propensity to restrain managerial opportunism. Nevertheless, these are preliminary results without the inclusion of control variables and therefore inferences can be drawn after the inclusion of the control variables in our main regression model. In general, the size of each correlation coefficient in Table 2 does not exceed the 0.80 threshold that has been documented in the existing literature to trigger any potential multicollinearity problems (Hair et al., 2006; Sharma et al., 2017; Ullah et al., 2022; Owusu et al., 2024; Ullah et al., 2024).

4.2. Multivariate analysis

Prior to testing the effects of monitoring and advisory lead independent director role on earnings management, we first use our sample to check the results of previous research. We, thus, estimate the effect of the mere presence of lead independent director representation on the board on earnings management using the following regression model:

$$DA_{it} = \alpha + \beta_1 LEAD_{it+k_{it}} + \beta_2 Controls_{it} + \alpha_j + \delta_t + \varepsilon_{it} \quad (3)$$

where the dependent variable DA is the Modified Jones discretionary

Table 1
Summary statistics.

Variable Name	Observations	Mean	Standard Deviation	Median	Minimum	Maximum
DA	6664	0.092	0.108	0.051	0.000	1.236
LMONT	6664	0.089	0.045	0.000	0.000	0.333
LADV	6664	0.038	0.027	0.000	0.000	0.200
SIZE	6664	8.704	1.516	8.782	-0.860	11.998
LEV	6664	0.308	0.091	0.304	0.000	0.717
OCF	6664	0.066	0.136	0.067	-5.910	1.751
ROA	6664	2.661	5.513	-0.636	-2.800	9.080
MBV	6664	6.469	8.785	3.271	1.118	25.677
SALESG	6664	0.043	0.253	0.033	-0.941	7.298
LOSS	6664	0.076	0.265	0.000	0.000	1.000
BDSIZE	6664	10.589	1.980	11.000	5.000	19.000
PctIndDir	6664	62.960	22.066	66.667	5.882	100.000
PctFemDir	6664	0.173	0.102	0.167	0.000	0.714
Duality	6664	0.392	0.490	0.000	0.000	1.000
Big4	6664	0.937	0.242	1.000	0.000	1.000
MONTSIZE	6664	4.29	2.31	4.000	0.000	9.000
ADVSIZE	6664	3.83	1.80	3.000	0.000	7.000
FINEXP	6664	21.211	15.952	16.712	0.072	0.687

Note: Variable definitions are available in Appendix.

Table 2
Pearson correlation matrix.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
(1) DA	1																	
(2) LMONT	-0.005	1																
(3) LADV	0.008	0.063	1															
(4) SIZE	0.106	-0.090	-0.006	1														
(5) LEV	0.010	-0.098	-0.066	0.227	1													
(6) OCF	-0.479	-0.003	0.017	-0.094	-0.013	1												
(7) ROA	-0.347	-0.018	0.003	0.026	0.091	0.104	1											
(8) MBV	0.135	0.069	0.049	0.202	-0.233	0.149	-0.245	1										
(9) SALESG	0.016	-0.041	-0.004	-0.090	-0.010	-0.069	-0.174	0.087	1									
(10) LOSS	-0.269	-0.054	-0.006	0.041	0.108	-0.113	0.381	-0.257	-0.115	1								
(11) BDSIZE	0.031	0.105	-0.008	0.480	0.106	-0.058	0.089	0.124	-0.071	-0.056	1							
(12) PctIndDir	-0.073	0.126	0.068	-0.009	-0.079	-0.052	0.081	-0.137	-0.049	-0.008	0.001	1						
(13) PctFemDir	-0.188	0.043	0.027	0.099	-0.172	-0.170	-0.026	0.207	-0.080	-0.076	-0.019	0.021	1					
(14) Duality	0.022	0.023	-0.008	-0.004	-0.010	-0.011	0.001	0.007	-0.013	-0.004	-0.023	0.011	0.096	1				
(15) Big4	-0.038	0.085	0.061	0.399	0.121	0.077	0.011	0.173	-0.038	-0.016	0.161	0.089	-0.002	-0.002	1			
(16) MONTSIZE	-0.018	0.251	-0.028	0.101	-0.014	-0.027	0.096	-0.102	-0.063	-0.051	0.252	0.487	-0.029	-0.026	0.201	1		
(17) ADVSIZE	0.070	-0.012	0.090	0.255	0.046	-0.051	0.045	-0.015	0.003	0.041	0.374	0.347	-0.131	0.017	0.048	-0.193	1	
(18) FINEXP	-0.070	-0.039	0.063	0.030	-0.005	0.001	0.021	0.129	0.003	-0.015	-0.151	-0.133	0.229	0.020	0.020	-0.127	-0.074	1

Note: Variable definitions are available in Appendix.

accruals, subindex i stands for firm, t stands for time, and $LEAD$ takes the value of 1 if a firm has a lead independent director representation on the board, and 0 otherwise. $Controls_{i,t}$ is a vector of control variables. In common with existing literature (Alhadab & Clacher, 2018; Owusu et al., 2022; Hsu et al., 2024), we include a number of firm and governance characteristics shown to impact earnings management. These control variables include firm size ($SIZE$), leverage (LEV), operating cash flows (OCF), return on assets (ROA), market-to-book value (MBV), sales growth ($SALESG$), loss ($LOSS$), board size ($BDSIZE$), percentage of independent directors ($PctIndDir$), percentage of female directors ($PctFemDir$), percentage of directors with financial expertise ($FINEXP$), CEO duality ($Duality$) and big four auditors ($Big4$). (α_j) is industry fixed effects and (δ_t) is year fixed effects. The Appendix contains all the variable definitions.

In Table 3, we present the regression results from equation (3) of how the mere presence of lead independent directors within corporate boardrooms affects earnings management. As Models 1 (without corporate governance control variables) and 2 (with all control variables) reveal, a lead independent director representation on the board has a negative impact on discretionary accruals. This finding confirms Bryan and Mason (2023) and Hsu et al. (2024) studies, but it is in sharp contrast with Larcker et al. (2007) finding of no association between lead independent directors and earnings management. Our finding is also economically meaningful, in that, the coefficient of $LEAD$ in Table 3 (0.0026) and the average discretionary accruals in Table 1 (0.092), lead to a decreasing effect of 2.83% (0.0026/0.092) in earnings management.

Next, we estimate the monitoring and advisory lead independent directors on earnings management using the following equation:

$$DA_{i,t} = \alpha + \beta_1 ROLE_{i,t+k_{i,t}} + \beta_2 Controls_{i,t} + \alpha_j + \delta_t + \varepsilon_{i,t} \quad (4)$$

Table 3

Lead independent director and earnings management.

Variables	Model (1) DA	(Model 2) DA
LEAD	-0.0026*** (-2.70)	-0.0026** (-2.11)
SIZE	-0.0042*** (-4.19)	-0.0041*** (-4.09)
LEV	0.0190 (1.28)	0.0203 (1.37)
OCF	0.1912*** (5.88)	0.1929*** (5.92)
ROA	-0.0001*** (-3.35)	-0.0001*** (-3.83)
MBV	0.0002*** (3.05)	0.0002*** (3.01)
SALESG	0.0021 (0.71)	0.0026 (0.87)
LOSS	0.0487*** (8.34)	0.0491*** (8.39)
BDSIZE	-	-0.0008** (-2.23)
PctIndDir	-	-0.0002*** (-3.21)
PctFemDir	-	-0.0365*** (-3.96)
Duality	-	0.0089*** (3.34)
Big4	-	-0.0119** (-1.96)
FINEXP	-	-0.0075** (-2.08)
Constant	0.041*** (5.11)	0.0624*** (3.96)
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Adjusted R-squared	0.0463	0.0975
Number of Observations	6664	6664

Note: Variable definitions are available in Appendix. t statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

except for *ROLE* and two additional control variables, all variables are defined under equation (3). As already highlighted in subsection 3.3, *ROLE* represents monitoring lead independent directors (*LMONT*) or advisory lead independent directors (*LADV*). Hence, the coefficient β_1 captures the direct effect of the monitoring and advisory lead independent directors. In line with our earlier discussion, we expect the coefficient of β_1 to be negative when *ROLE* is monitoring ($k = \text{monitoring}$). Alternatively, β_1 can be insignificant if *ROLE* is advisory ($k = \text{advisory}$). The two additional control variables included in equation (4) are the number of monitoring directors (*MONTSIZE*) and the number of advisory directors (*ADVSIZE*).

Models 1 and 2 of Table 4 show our main results of the effects of monitoring and advisory lead independent directors on earnings management. In Model 1, we regress *DA* on *LMONT* and, the coefficient on *LMONT* is -0.0045 ($t\text{-statistics} = -3.27$). This suggests that a lead independent director who plays a monitoring role reduces earnings management. In model 2, we examine whether a lead independent director with an advisory role has varying effects on earnings management. The coefficient on *LADV* is negative but not statistically significant. We hypothesised that lead independent directors who sit on the independent directors-dominated monitoring committees are more likely to restrain managerial opportunism than those who sit on the executive-dominated advisory committees. Our results in Models 1 and 2 of Table 4 are consistent with our expectation, in that, monitoring lead independent directors rather than advisory lead independent directors have the propensity to restrain managers from engaging in earnings management. Thus, our hypothesis 1 is supported. Our results are also economically significant, in that, the coefficient of *LMONT* in Model 1 of Table 4 (0.0045) and the average discretionary accruals in Table 1 (0.092), lead to a decreasing effect of 4.89% (0.0045/0.092) in earnings management.

4.3. Robustness

Thus far, the baseline analysis suggests that monitoring lead independent directors rather than advisory lead independent directors' constraint managerial opportunism. We undertake further analysis to check the robustness of our baseline results. First, we employ Kothari et al. (2005) performance-matched model by including *ROA* in equation (1) to estimate an alternative dependent variable of performance-matched discretionary accruals (*PMDA*). The results are contained in Models 3 and 4 of Table 4. In Models 3 and 4, *LMONT* rather than *LADV* is negatively significant with *PMDA*, implying that our baseline results are not sensitive to an alternative measure of earnings management. Second, we replace *LMONT* with the percentage of monitoring lead independent directors relative to the total number of board members (*LMONT1*) and *LADV* with the percentage of advisory lead independent directors relative to the total number of board members (*LADV1*) as alternative measures and the results are presented in Models 5 and 6 of Table 4. Even though the coefficients on *LMONT1* and *LADV1* have become smaller, the results consistently show that *LMONT1* rather than *LADV1* is negative and significantly associated with *DA*.

Third, accounting literature (e.g., Gunny, 2010; Alhadab & Clacher, 2018) suggests that previous year earnings management affects a firm's current year earnings manipulative behaviour. For that reason, we control the effects of the previous year's earnings management. Table 5 presents the results. In Model 1, the coefficient on *LMONT* is negative and statistically significant at the 1% level, whilst in Model 2, the coefficient on *LADV* is negative but not statistically significant. These results show that our baseline results are not sensitive to the previous year earnings management.

Prior research (i.e., Faleye et al., 2011) suggests that monitoring (advising) quality improves (weakens) when the majority of independent directors serve on monitoring committees. In our fourth sensitivity test, we group our sample by the size (i.e., high vs low) of independent

directors that serve on monitoring (advisory) committees using the median of *MONTSIZE* (*ADVSIZE*) and the results are reported in Table 6. As Models 1 and 3 of Table 6 show, the results continue to be negative and statistically significant at the 5% level or higher but it is more pronounced in firms with low intensive monitoring independent directors. In a similar vein, the results in Models 2 and 4 of Table 6 continue to be negative but not statistically significant. Therefore, we conclude that our baseline results are not sensitive to intensive monitoring/advisory independent directors.

4.4. Addressing endogeneity

It is well documented in the empirical accounting literature that studies like ours may suffer from endogeneity concerns due to self-selection bias, reverse causality and omitted variables bias (Larcker & Rusticus, 2010; Gippel et al., 2015). For these reasons, the conclusions from our baseline analysis may be misleading. We, thus, address the endogeneity concerns by implementing different research designs including the propensity score matching (PSM), Heckman two-step and two-stage least squares (2SLS) that have been widely used in the existing literature.

First, we use the PSM method to mitigate the concerns about potential endogeneity issues. We first compare the earnings management practices between a group of firms with lead independent directors that have a monitoring role (*LMONT*) and a matched group with lead independent directors that do not have a monitoring role on the boards. In common with Hsu et al. (2024), we use the nearest neighbour matching approach with replacement to match each firm with *LMONT* and without *LMONT*. Similarly, we repeat the PSM procedures for firms with lead independent directors that have an advisory role (*LADV*) and match them with the nearest-neighbour firm without *LADV*. We then re-estimate equation (4) using the matched sample firms and report the results in Table 7. As Models 1 and 2 of Table 7 indicate, while the coefficient on *LMONT* continues to be negatively and significantly related to discretionary accruals, the coefficient on *LADV* is negative but not statistically significant. These findings support the conclusions drawn from our baseline analysis and appear not to be affected by endogeneity issues.

Second, according to Antonakis et al. (2010) and Larcker and Rusticus (2010), an alternative research design is needed to address endogeneity when undertaking a study of this nature. We, therefore, employ an alternative two-step model suggested by Heckman (1976) to address endogeneity issues. First, we run a probit regression to check the possibility of firms appointing a monitoring or an advisory lead independent director. Specifically, we include the industry average of the percentage of lead independent directors (*IndPctLead*) in addition to all the control variables in the probit model in the first step. We then use the estimation from the first step to compute the inverse Mills ratio (*MILLS*) and include it in the second step as a control variable in equation (4). Our reported results from Models 3 and 4 of Table 7 support our baseline analysis, indicating that the conclusions drawn from our baseline analysis appear not to be affected by endogeneity issues.

Finally, to address any endogeneity concerns that may arise due to reverse causality, we employ the 2SLS instrumental variables method. Specifically, we use *IndPctLead* in the first stage as our instrumental variable. Theoretically, this instrument has the propensity to influence the appointment of a monitoring or an advisory lead independent director, but it is less likely to affect managerial opportunism (*DA*). In the second stage, we use the predicted *LMONT* and *LADV* to explain managerial opportunism. As Model 1 of Table 8 indicate, *LMONT* continues to be negative and significantly related to *DA*. In Model 2 of Table 8, *LADV* is still negative and not significantly related to *DA*. We also follow Larcker and Rusticus (2010) and undertake the Durbin-Wu-Hausman test for endogeneity but all the p-values from the regressions in both Models 1 and 2 of Table 8 are higher than 10% significance level, supporting the null hypothesis of no endogeneity. These results show

Table 4
Monitoring versus advisory lead independent directors and earnings management.

Variables	Model (1) DA	Model (2) DA	Model (3) PMDA	Model (4) PMDA	Model (5) DA	Model (6) DA
LMONT	−0.0045*** (−3.27)	—	−0.0041*** (−3.58)	—	—	—
LADV	—	−0.0028 (−1.63)	—	−0.0022 (−1.45)	—	—
LMONT1	—	—	—	—	−0.0002*** (−2.92)	—
LADV1	—	—	—	—	—	0.0000 (0.19)
SIZE	−0.0042*** (−4.19)	−0.0041*** (−4.09)	−0.0015*** (−5.30)	−0.0015*** (−5.37)	−0.0019*** (−2.93)	−0.0017*** (−2.73)
LEV	0.0190 (1.28)	0.0203 (1.37)	0.0046 (0.86)	0.0047 (0.88)	0.0261 (1.62)	0.0063 (1.44)
OCF	0.1912*** (5.88)	0.1929*** (5.92)	0.1494*** (5.36)	0.1497*** (5.37)	0.1847*** (5.85)	0.2115*** (6.81)
ROA	−0.0001*** (−3.35)	−0.0001*** (−3.83)	−0.1824*** (−4.79)	−0.1825*** (−4.80)	−0.0001** (−2.54)	−0.0001*** (−3.47)
MBV	0.0002*** (3.05)	0.0002*** (3.01)	0.0001*** (5.82)	0.0001*** (5.69)	0.0003*** (4.43)	0.0002** (2.44)
SALESG	0.0021 (0.71)	0.0026 (0.87)	0.0001 (1.21)	0.0001 (1.17)	0.0045 (0.23)	0.0046 (0.84)
LOSS	0.0487*** (8.34)	0.0491*** (8.39)	0.0231*** (8.39)	0.0232*** (8.41)	0.0379*** (12.15)	0.0448*** (10.02)
BDSIZE	−0.0008** (−2.23)	−0.0008** (−2.19)	−0.0011*** (−4.06)	−0.0011*** (−4.04)	−0.0011** (−2.01)	−0.0009** (−2.08)
PctIndDir	−0.0002*** (−3.21)	−0.0002*** (−3.12)	−0.0003** (−2.51)	−0.0003** (−2.42)	−0.0001** (−2.57)	−0.0002*** (−3.86)
PctFemDir	−0.0365*** (−3.96)	−0.0358*** (−3.88)	−0.0116*** (−3.28)	−0.0115*** (−3.26)	−0.0352*** (−4.84)	−0.0258*** (−3.16)
Duality	0.0089*** (3.34)	0.0082*** (3.11)	0.0021** (2.53)	0.0023*** (2.63)	0.0080** (2.33)	0.0077** (2.01)
Big4	−0.0119** (−1.96)	−0.0119** (−1.97)	−0.0035** (−2.10)	−0.0036** (−2.17)	−0.0031* (−1.77)	−0.0137*** (−2.81)
FINEXP	−0.0073** (−1.99)	−0.0078** (−2.09)	−0.0086** (−2.05)	−0.0098** (−2.15)	−0.0018*** (−4.18)	−0.0015*** (−3.76)
MONTSIZE	−0.0015*** (−3.70)	−0.0018*** (−4.33)	−0.0021*** (−2.61)	−0.0013** (−2.20)	−0.0021*** (−4.29)	−0.0016*** (−3.53)
ADVSIZE	0.0118*** (3.81)	0.0118*** (3.83)	0.0005** (2.41)	0.0005** (2.42)	0.0766* (1.67)	0.0788* (1.72)
Constant	0.0624*** (3.82)	0.0618*** (3.78)	0.0357*** (8.90)	0.0359*** (8.95)	0.0252*** (3.34)	0.0487*** (4.13)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R-squared	0.0981	0.0975	0.1741	0.1737	0.0840	0.0932
Number of Observations	6,664	6,664	6,664	6,664	6,664	6,664

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

that having *LMONT* results in less earnings management.

4.5. Monitoring versus advisory lead independent directors and earnings persistence

In this section, we examine the consequences of earnings persistence around monitoring and advisory lead independent directors. While our main results suggest that monitoring rather than advisory lead independent directors are associated with lower earnings management, [Hsu and Hu \(2016\)](#) argued and provided evidence to show that firms with independent directors in advisory roles rather than monitoring roles are associated with more earnings persistence and higher future cash flows. This contrasting evidence suggests that monitoring and advisory lead independent directors may play complementary roles when it comes to the overall corporate performance. As an additional test, we follow previous literature ([Barth et al., 2001](#); [Sloan, 1996](#); [Richardson et al.,](#)

[2005](#); [Hsu & Hu, 2016](#)) and utilise earnings persistence measures including, (1) earnings (*Earn*) defined as net income before extraordinary items scaled by average total assets, (2) cash flows (*CFO*) defined as cash flows from operating activities scaled by average total assets, and (3) accruals (*ACC*) defined as the difference between net income before extraordinary items and cash flows from operating activities scaled by average total assets to proxy earnings quality.

Panels A to C of [Table 9](#) present the results from the regressions for future earnings and cash flows on current earnings. As in [Hsu and Hu \(2016\)](#), we utilise future 1-year period, 2-year period and 3-year period reported earnings and cash flows from operations as our dependent variables. As Panel A shows, the coefficient on *LMONT* × *Earn* is only significant in Model 5 at the 10% level and is not significant in all other models. However, Models 1 to 6 of Panel B reveal that the coefficients on *LADV* × *Earn* are positive and statistically significant at the 5% level or higher, suggesting that the presence of advisory lead independent

Table 5
Controlling for previous year earnings management.

	Model (1) DA	Model (2) DA
LMONT	−0.0023*** (−3.45)	—
LADV	—	−0.0018 (−1.59)
DA _{t-1}	0.0807*** (2.82)	0.0801*** (2.80)
SIZE	−0.0014** (−2.06)	−0.0014** (−2.08)
LEV	0.0182 (1.28)	0.0192 (1.35)
OCF	0.1894*** (6.15)	0.1902*** (6.17)
ROA	−0.0001*** (−3.77)	−0.0001*** (−3.82)
MBV	0.0002*** (3.64)	0.0002*** (3.58)
SALESG	0.0035 (0.68)	0.0038 (0.77)
LOSS	0.0424*** (10.79)	0.0426*** (10.83)
BDSIZE	−0.0006** (−2.52)	−0.0006** (−2.53)
PctIndDir	−0.0002*** (−3.72)	−0.0002*** (−3.60)
PctFemDir	−0.0357*** (−4.26)	−0.0357*** (−4.25)
Duality	0.0081*** (2.02)	0.0082*** (2.07)
Big4	−0.0135*** (−2.84)	−0.0134*** (−2.82)
FINEXP	−0.0071** (−1.99)	−0.0079** (−2.11)
MONTSIZE	−0.0014*** (−3.36)	−0.0017*** (−4.01)
ADVSIZE	0.00116*** (3.55)	0.0117*** (3.57)
Constant	0.0403*** (3.99)	0.0403*** (3.99)
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Adjusted R-squared	0.1128	0.1166
Number of Observations	6,664	6,664

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

directors improves earnings persistence and future cash flows. In a combined regression where $LMONT \times Earn$ and $LADV \times Earn$ are included to test the incremental effects of monitoring and advisory lead independent directors, our results in Models 1 to 6 in Panel C show that the coefficients on $LADV \times Earn$ rather than $LMONT \times Earn$ continue to be positive and statistically significant, confirming our earlier results that monitoring (advisory) lead independent directors are less (more) likely to improve earnings persistence and future cash flows.

In Table 10, we decompose earnings to capture cash flows and accruals components, and the results are detailed in Panels A and B for monitoring intensity and advisory intensity, respectively. While the coefficients on $LMONT \times CFO$ and $LMONT \times ACC$ are not significant in Panel A except for Model 1 where the coefficient on $LMONT \times CFO$ is only significant at the 10% level, the coefficients on $LADV \times CFO$ and $LADV \times ACC$ are consistently positive and significant at the 5% level or higher in Panel B. We unveil that both cash flows and current accruals components consistently improve future earnings in firms with advisory lead independent directors. These results imply that both regular independent directors' monitoring and advisory role measures used in Hsu

Table 6
Analysis of high versus low monitoring/advisory intensive directors.

	(1) High monitoring	(2) High advisory	(3) Low monitoring	(4) Low advisory
LMONT	−0.0044** (−1.99)	—	−0.0052*** (−2.73)	—
LADV	—	−0.0021 (−0.97)	—	−0.033 (−0.77)
SIZE	−0.0019*** (−2.88)	−0.0014** (−2.03)	−0.0014*** (−3.03)	−0.0014*** (−3.07)
LEV	0.0418* (1.79)	0.0115 (0.94)	−0.0212 (−1.53)	−0.0209 (−1.52)
OCF	0.2060*** (5.80)	0.1754*** (2.95)	0.2029*** (7.27)	0.2031*** (7.28)
ROA	−0.0001*** (−5.81)	−0.0001*** (−3.20)	0.0001 (0.36)	0.0001 (0.23)
MBV	0.0002** (2.11)	0.0002** (2.30)	−0.0001 (−1.59)	−0.0001 (−1.60)
SALESG	0.0066 (1.49)	0.0067 (0.76)	0.0024 (0.38)	0.015 (0.46)
LOSS	0.0470*** (6.80)	0.0263*** (5.09)	0.0401*** (9.90)	0.0402*** (9.91)
BDSIZE	−0.0020*** (−2.64)	−0.0023** (−2.52)	−0.0012** (−2.56)	−0.0014** (−2.57)
PctIndDir	−0.0004*** (−2.69)	−0.0003*** (−2.64)	−0.0001** (−2.51)	−0.0002** (−2.40)
PctFemDir	−0.0088*** (−2.84)	−0.0034** (−2.13)	−0.0003 (−0.053)	−0.0004 (−0.07)
Duality	0.0047** (2.06)	0.0036** (2.49)	0.0001 (0.25)	−0.0003 (−0.79)
Big4	−0.0138** (−2.28)	−0.0032* (−1.81)	−0.0104** (−2.38)	−0.0012** (−2.36)
FINEXP	−0.0079** (−2.21)	−0.0083** (−1.98)	−0.0026 (−0.33)	−0.0015 (−0.88)
MONTSIZE	—	−0.0015* (−1.90)	—	0.0003 (0.86)
ADVSIZE	0.0210*** (2.86)	—	0.0003 (0.75)	—
Constant	0.0724*** (3.7440)	0.0349*** (3.5749)	0.0556*** (4.7850)	0.0557*** (4.8454)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.4617	0.5732	0.5055	0.5058
Number of Observations	3252	3286	3412	3378

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and Hu (2016) and the lead independent directors' monitoring and advisory role measures utilised in this study provide insights and deepen our understanding of the earnings quality effect of board monitoring and advisory roles.

Taken together, our findings highlight that while lead independent directors with monitoring rather than advisory roles have the propensity to reduce abnormal accruals, those in advisory rather than monitoring roles improve earnings persistence and future cash flows. Therefore, we can conclude that both lead independent directors' monitoring and advisory roles contribute to corporate performance in complementary ways.

4.6. Lead independent director roles, financial expertise, and earnings management

Lead independent directors are to ensure a balance between managerial autonomy with the necessary monitoring. In performing their

Table 7
Propensity score matching (PSM) and Heckman two-stage model.

Variables	PSM		Heckman	
	Model (1)	Model (2)	Model (3)	Model (4)
LMONT	DA −0.0048** (−2.15)	DA — (−0.96)	DA −0.0068** (−2.05)	DA — (−0.78)
LADV	— — —	−0.0013 (−0.96)	— (−0.78)	−0.0032 (−0.78)
MILLS	— — —	— — —	0.7529*** (3.36)	0.6806*** (3.06)
SIZE	−0.0018*** (−2.61)	−0.0024** (−2.58)	−0.0019*** (−2.59)	−0.0015*** (−2.52)
LEV	0.0156 (1.08)	0.0154 (1.33)	0.0158 (1.01)	0.0153 (1.34)
OCF	0.1887** (2.17)	0.2014** (2.48)	0.2057*** (7.37)	0.2092** (2.02)
ROA	−0.0004*** (−3.97)	−0.0004*** (−2.80)	−0.0002** (−2.31)	−0.0003** (−2.06)
MBV	0.0003** (2.50)	0.0003** (2.21)	0.0002** (2.33)	0.0002** (2.14)
SALESG	0.0025 (1.20)	0.0024 (1.40)	0.0051 (1.15)	0.0038 (1.10)
LOSS	0.0127** (2.49)	0.0246*** (2.79)	0.0137*** (2.67)	0.0298*** (3.18)
BDSIZE	−0.0011** (−2.10)	−0.0019*** (−2.60)	−0.0006** (−2.28)	−0.0007** (−2.32)
PctIndDir	−0.0001*** (−2.87)	−0.0001*** (−2.86)	−0.0003** (−2.34)	−0.0003*** (−3.09)
PctFemDir	−0.0038** (−2.32)	−0.0056** (−2.53)	−0.0358** (−2.39)	−0.0363*** (−3.06)
Duality	0.0066** (2.52)	0.0070*** (2.68)	0.0081*** (3.03)	0.0083*** (3.01)
Big4	−0.0174** (−2.01)	−0.0181** (−2.53)	−0.0133** (−2.26)	−0.0132** (−2.08)
FINEXP	−0.0076** (−2.44)	−0.0081** (−2.26)	−0.0067*** (−2.64)	−0.0062*** (−2.64)
MONTSIZE	−0.0029** (−2.22)	−0.0031** (−2.11)	0.0015** (2.37)	0.0016** (2.08)
ADVSIZE	0.0011* (1.74)	0.0030** (2.0127)	−0.0117** (−2.35)	−0.0116 (−0.05)
Constant	0.0502*** (3.70)	0.0263 (1.25)	−2.1931** (−2.33)	0.1874 (1.09)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.0637	0.0811	0.5071	0.5070
Number of Observations	1199	533	6,664	6,664

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

duties, these directors require financial knowledge in order to understand the financial reporting process and restrict managerial opportunism. If the monitoring lead independent directors than advisory lead independent directors exert influences on their firm to reduce earnings management, then their impact on managerial opportunism should be more prominent in firms with monitoring lead independent directors with financial expertise than in other firms. This new analysis differs from previous studies (e.g., Bryan & Mason, 2023; Hsu et al., 2024) that focus on lead independent directors' experience and expertise such as their audit committee membership, age, tenure and higher educational background. We, thus, test our hypothesis 2 in this section.

As noted earlier, we interact *ROLE* (*LMONT* and *LADV*) with *LEADFE* and test their effect on managerial opportunism. In Model 1 of Table 11, the results show that the association between the interaction term, *LMONT* × *LEADFE*, is more pronounced relative to the baseline analysis.

Table 8
2SLS regression.

Variables	First stage		Second stage	
	Model (1)	Model (2)	Model 3	Model 4
	LMONT	LADV	DA	DA
LMONT	—	—	−0.0056** (−2.03)	—
LADV	—	—	—	−0.0027 (−1.61)
IndPctLead	0.2683*** (7.84)	0.0967*** (5.30)	—	—
SIZE	−0.0033*** (−4.61)	−0.0011*** (−3.41)	−0.0057*** (−5.04)	−0.0050*** (−5.55)
LEV	−0.0329*** (−4.90)	−0.0137*** (−3.30)	−0.0393 (−1.40)	−0.0354 (−1.41)
OCF	−0.0351* (−1.95)	0.0187** (1.97)	0.1877*** (8.83)	0.2241*** (10.30)
ROA	0.0002*** (5.09)	−0.0003 (−0.77)	−0.0001*** (−2.66)	−0.0001*** (−3.27)
MBV	−0.0012 (−1.10)	−0.0018** (−2.07)	0.0002*** (3.87)	0.0002*** (4.17)
SALESG	−0.0080*** (−3.39)	0.0099*** (4.17)	0.0064 (0.73)	0.0041 (0.40)
LOSS	−0.0088*** (−4.60)	−0.0009 (−0.80)	0.0389*** (10.99)	0.0426*** (15.50)
BDSIZE	0.0001 (0.27)	−0.0009*** (−3.45)	−0.0007** (−2.44)	−0.0006** (−2.42)
PctIndDir	−0.0001* (−1.91)	0.0001*** (2.82)	−0.0004** (−2.45)	0.0004** (2.56)
PctFemDir	0.0205*** (3.21)	0.0175*** (4.25)	−0.0279*** (−2.92)	−0.0342*** (−3.26)
Duality	0.0174*** (11.67)	0.0061*** (8.01)	0.0161*** (3.30)	0.0130*** (3.75)
Big4	0.0071** (2.39)	0.0079*** (6.21)	−0.0023*** (−2.62)	−0.0016** (−2.30)
FINEXP	0.0140*** (3.1028)	0.0112*** (4.5145)	−0.0128** (−2.47)	−0.0159** (−2.56)
MONTSIZE	0.0062*** (13.65)	−0.0004* (−1.66)	−0.0030** (−1.96)	−0.0026** (−2.19)
ADVSIZE	0.0014*** (3.50)	0.0013*** (5.29)	0.0110** (1.99)	0.0115* (1.86)
Constant	0.0069 (1.23)	0.0132*** (4.23)	0.0680*** (9.59)	0.0772*** (7.92)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Adjusted R-squared	0.1086	0.0371	0.4379	0.2178
Number of Observations	6,664	6,664	6,664	6,664

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Interestingly, the association between the interaction term, *LADV* × *LEADFE*, is also negative and statistically significant at the 1% level in Model 2 of Table 11. These results reveal that, regardless of the role that they play, lead independent directors with financial expertise reduce managerial opportunism.²

5. Conclusion

As the move to implement the lead independent director policy has intensified, policymakers and investors have the incentive to know the role play by lead independent directors in managerial opportunism. So far, extant literature has focused on the effect of the mere participation of lead independent directors on managerial opportunism, paying less attention to how lead independent directors' roles within boardrooms may affect managerial opportunism, notably, their monitoring and advisory roles. We fill this gap by examining the effect of monitoring and

² As in our baseline analysis, we address endogeneity concerns using PSM and Heckman two-step model and the results (untabulated) are qualitatively similar to those reported in Table 11.

Table 9
Monitoring versus advisory lead independent directors and earnings persistence using cash flows and earnings.

Panel A: Monitoring intensity						
	(1)	(2)	(3)	(4)	(5)	(6)
	Earn _{t+1}	CFO _{t+1}	Earn _{t+2}	CFO _{t+2}	Earn _{t+3}	CFO _{t+3}
Earn _t	0.2088*** (2.83)	0.1256*** (4.56)	0.0366*** (2.97)	0.0253** (2.51)	0.1212*** (3.76)	0.0394*** (2.69)
LMONT _t	0.0094 (0.53)	−0.0406** (−2.30)	0.0015 (0.72)	−0.0638* (−1.91)	0.2103 (1.3877)	−0.1634*** (−3.76)
LMONT _t × Earn _t	0.7954 (1.56)	0.3262 (1.34)	−0.1815 (−0.1243)	0.1890 (0.34)	0.1284* (1.71)	0.1682 (1.43)
Constant	−0.0185*** (−2.75)	0.0365*** (7.21)	−0.0113 (−0.0225)	0.0327*** (5.0123)	0.0235** (2.0883)	0.0299*** (4.4660)
Controls included	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R squared	0.2195	0.3608	0.2279	0.4181	0.2334	0.3625
Number of Observations	5,451	5,451	4,441	4,441	3,610	3,610
Panel B: Advisory intensity						
	(1)	(2)	(3)	(4)	(5)	(6)
	Earn _{t+1}	CFO _{t+1}	Earn _{t+2}	CFO _{t+2}	Earn _{t+3}	CFO _{t+3}
Earn _t	0.2228*** (3.36)	0.0855*** (3.33)	0.0189** (3.14)	0.0678** (2.01)	0.1727** (2.20)	0.1747*** (4.14)
LADV _t	−0.0214 (−0.86)	−0.1093*** (−3.91)	−0.0270 (−1.02)	−0.0870** (−2.43)	−0.0169 (−1.09)	−0.0343** (−2.11)
LADV _t × Earn _t	0.0460** (2.23)	0.1451*** (3.59)	0.4093*** (4.56)	0.0335*** (−2.70)	0.0919*** (4.97)	0.0261*** (2.66)
Constant	0.0197*** (2.93)	0.0398*** (7.77)	0.0911*** (3.09)	0.0322*** (4.89)	0.0233** (2.06)	0.0303*** (4.51)
Controls included	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R squared	0.2189	0.3542	0.2366	0.4196	0.2345	0.3620
Number of Observations	5,451	5,451	4,441	4,441	3,610	3,610
Panel C: Monitoring and advisory intensity						
	(1)	(2)	(3)	(4)	(5)	(6)
	Earn _{t+1}	CFO _{t+1}	Earn _{t+2}	CFO _{t+2}	Earn _{t+3}	CFO _{t+3}
Earn _t	0.2081*** (2.81)	0.1233*** (4.54)	2.2229** (2.38)	0.1809*** (4.13)	0.2492** (2.06)	0.1994*** (4.46)
LMONT _t	0.0123 (0.68)	0.0377** (2.15)	0.0238 (0.61)	−0.0206 (−1.20)	0.1035*** (3.33)	−0.0314 (−1.29)
LADV _t	−0.0225 (−0.90)	−0.1042*** (−3.78)	−0.2608*** (−4.27)	−0.1029*** (−2.96)	−0.0236 (−0.59)	−0.1236*** (−4.79)
LMONT _t × Earn _t	0.8350 (1.62)	−0.2569 (−1.23)	0.2163 (0.144)	0.0915 (0.16)	−1.4777 (−1.45)	0.9804 (1.15)
LADV _t × Earn _t	0.1379** (2.44)	0.7664*** (3.19)	0.9323*** (4.80)	0.4430*** (3.13)	0.4123*** (5.63)	0.5345** (2.45)
Constant	0.0187*** (2.76)	0.0373*** (7.35)	0.0165** (2.59)	0.0403*** (6.26)	0.0142** (2.34)	0.0414*** (6.37)
Controls included	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R squared	0.2205	0.3635	0.2290	0.4085	0.2243	0.3458
Number of Observations	5,451	5,451	4,441	4,441	3,610	3610

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10

Monitoring versus advisory lead independent directors and earnings persistence using cash flows and accruals components.

Panel A: Monitoring intensity			
	(1)	(2)	(3)
	Earn _{t+1}	Earn _{t+2}	Earn _{t+3}
CFO _t	0.2530*** (3.66)	0.4572*** (4.96)	0.3169*** (2.80)
ACC _t	0.2115*** (2.89)	0.1164 (3.28)	0.1983*** (3.30)
LMONT _t	−0.0416 (−1.1641)	0.0989 (1.3685)	−0.0203 (−0.4669)
LMONT _t × CFO _t	0.1499* (1.92)	−0.1523 (−0.11)	−0.3414 (−1.29)
LMONT _t × ACC _t	0.3357 (0.52)	−0.7541 (−1.08)	−0.1245 (−1.01)
Constant	−0.0111* (−1.68)	0.0034 (0.32)	0.0153 (1.41)
Controls included	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Adjusted R squared	0.2188	0.2165	0.2238
Number of Observations	5,451	4,441	3,610
Panel B: Advisory intensity			
	(1)	(2)	(3)
	Earn _{t+1}	Earn _{t+2}	Earn _{t+3}
CFO _t	0.2294*** (3.88)	0.5006*** (5.71)	0.3499*** (3.26)
ACC _t	0.2229*** (3.36)	0.0230** (2.40)	0.1485*** (3.03)
LADV _t	−0.0137 (−0.37)	−0.0165 (−0.66)	−0.044 (−0.75)
LADV _t × CFO _t	0.0278** (2.52)	0.0796*** (5.05)	0.0714*** (5.16)
LADV _t × ACC _t	0.0518*** (2.71)	0.1548*** (4.14)	0.3307* (1.81)
Constant	−0.0132** (−2.06)	0.0047 (0.43)	0.0128* (1.79)
Controls included	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Adjusted R squared	0.2182	0.2256	0.2240
Number of Observations	5,451	4,441	3,610

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

advisory lead independent directors on managerial opportunism, and whether their financial expertise can make a difference.

Our analysis provides strong evidence that monitoring lead independent directors rather than advisory lead independent directors are negatively associated with discretionary accruals. We also find that advisory rather than monitoring lead independent directors improve earnings persistence and future cash flows. When we interact monitoring or advisory lead independent directors with financial expertise, our results show that, regardless of the role that they play, monitoring and advisory lead independent directors have a decreasing effect on managerial opportunism. Our results are robust to an alternative measure of earnings management, monitoring and advisory lead independent directors, and alternative research designs including PSM, Heckman two-step model, and 2SLS regression model.

Our findings make several contributions to the existing literature. First, previous literature looks at the impact of the mere presence of lead independent directors on firms' outcomes, and not much attention has been given to their role within corporate boardrooms. Considering that a lead independent director policy was introduced in the US to avoid the separation of the CEO and the board chair positions, we examine the role

Table 11

The role of financial expertise.

Variables	Model (1) DA	Model (2) DA
LMONT	−0.0056*** (−3.13)	—
LEADFE	−0.0081 (−1.21)	−0.0075* (−1.94)
LMONT × LEADFE	−0.0159*** (−3.06)	—
LADV	—	−0.0199 (−1.58)
LADV × LEADFE	—	−0.0086*** (−2.79)
SIZE	−0.0014** (−2.14)	−0.0014** (−2.20)
LEV	0.0208 (1.46)	0.0211 (1.48)
OCF	0.2071*** (6.76)	0.2063*** (6.74)
ROA	−0.0001*** (−4.14)	−0.0001*** (−4.14)
MBV	0.0002*** (3.62)	0.0002*** (3.60)
SALESG	0.0033 (0.67)	0.0033 (0.68)
LOSS	0.0452*** (10.12)	0.0453*** (10.14)
BDSIZE	−0.0006** (−2.44)	−0.0005** (−2.41)
PctIndDir	−0.0002*** (−3.95)	−0.0002*** (−3.95)
PctFemDir	−0.0365*** (−4.42)	−0.0363*** (−4.41)
Duality	0.0073** (2.09)	0.0078** (2.09)
Big4	−0.0143*** (−2.90)	−0.0142*** (−2.88)
MONTSIZE	−0.0018*** (−4.26)	−0.0018*** (−4.28)
ADVSIZE	0.0118*** (3.93)	−0.0118*** (−3.90)
Constant	0.0453*** (3.96)	0.0456*** (3.99)
Industry fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Adjusted R squared	0.0990	0.0990
Number of Observations	6,664	6,664

Note: Variable definitions are available in Appendix. *t* statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(monitoring and advisory) that lead independent directors can play to reduce managerial opportunism. Second, we expand the existing literature that focuses on monitoring and advisory roles of regular independent directors in earnings quality/persistence. We show that lead independent directors' monitoring and advisory roles contribute to corporate performance in complementary ways. Finally, our findings have important policy implications. Given that monitoring and advisory lead independent directors play complementary roles, it is important for firms to appoint a lead independent director into a monitoring role to ensure strong oversight, accountability and compliance, especially when it comes to management discipline around the quality of financial report, while in the areas of strategic insight and persistent earnings, an advisory lead independent director is suitable.

Our study is not without some limitations. For instance, our sample is from the US where a lead independent director policy was introduced to avoid the separation of the CEO and board chair positions. For this

reason, we caution against generalising our results to countries with different corporate governance environments. We would encourage future research to be undertaken in countries such as the UK where the combined role of the CEO and the board chair is not permitted but the lead independent director policy has been implemented with a different rationale. In addition, this study relied on larger firms that have the resources to appoint lead independent directors relative to the smaller firms. Therefore, caution should be exercised when generalising our results to smaller companies in the US or other jurisdictions. Moreover, our measure of managerial opportunism is based on discretionary accruals, as we are interested in the monitoring or advisory role of the lead

independent director around financial reporting decisions that violate the underlying accounting principles. Nevertheless, future research could also consider real activities earnings management that relates to opportunistic business decisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Variable definitions

	Definitions
Dependent variable	
DA	Modified Jones discretionary accruals estimated using Dechow et al. (1995) model.
Independent variables	
LEAD	Dummy variable coded 1 if a firm has a lead independent director on the board, and 0 otherwise.
LMONT	Percentage of lead independent monitoring directors to the number of independent monitoring directors.
LADV	Percentage of lead independent advisory directors to the number of independent advisory directors.
Moderation variable	
LEADFE	Dummy variable coded 1 if a lead independent committee role name has financial expert, and 0 otherwise.
Corporate governance variables	
BDSIZE	The total number of directors.
PctIndDir	Percentage of independent directors to the total number of directors.
PctFemDir	Percentage of female directors to the total number of directors.
Duality	Dummy variable coded 1 if the CEO is the chairman positions are occupied by one individual, and 0 otherwise.
Big4	Dummy variable coded 1 if the firm is audited by one of the big four audit firms, and zero otherwise.
FINEXP	Percentage of board of directors with financial expertise to the total number of directors.
MONTSIZE	Total number of independent monitoring directors.
ADVSIZE	Total number of independent advisory directors.
Control variables	
MBV	The proportion of firms' market value to the book value of common equity
SIZE	The natural logarithm of a firm's total asset.
LEV	The total debt to total asset.
OCF	Cash flows from operations to total assets
ROA	Net income before extraordinary items divided by total asset
SALESG	The change in the total sales of the firm
LOSS	Dummy variable coded 1 if net income before extraordinary items is a loss, and zero otherwise.

Data availability

Data will be made available on request.

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